W-P-796B
April 24, 1974
SUPERSEDING
Fed. Spec. W-P-796a
May 13, 1960

FEDERAL SPECIFICATION

PULLERS, FUSE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers pullers for use in removing and inserting cartridge-type fuses, 1/4 inch diameter to 2-1/2 inches diameter in size, that are used in electrical circuits up to 600 volts.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity indicated by the title of the specification or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification. Fuse pullers shall be of the following types and sizes as specified (see 6.2):

Type I - Laminated construction.
  Size 1 - 1/4 inch through 1/2 inch, fuse diameter range.
  Size 2 - 1/2 inch through 1-1/4 inches, fuse diameter range.
  Size 3 - 1 inch through 2-1/2 inches, fuse diameter range.

Type II - Molded construction.
  Size 1 - 1/4 inch through 1/2 inch, fuse diameter range.
  Size 2 - 1/2 inch through 1 inch, fuse diameter range.
  Size 3 - 1-1/2 inches through 1-3/4 inches, fuse diameter range.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-40 - Hand Tools, Packaging, Packing, and Marking of.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Design. Fuse pullers in accordance with this specification shall be suitable for removing (and inserting) cartridge type fuses used in electrical circuits up to 600 volts. The pullers shall be of the type and size specified (see 1.2 and 6.2) and shall be similar to the puller shown in figure 1 or figure 2, as applicable. The fuse-gripping surfaces shall provide a firm grip on the fuse body during insertion and removal from the fuse holders. Size 1 pullers shall have fuse-gripping surfaces on one end only. The surfaces of size 2 and size 3 pullers which are held in the hand shall be provided with a means to prevent slipping.

3.1.1 Type I laminated construction type. Type I pullers shall be similar to figure 1 and comply with the dimensions of table I. They shall be fabricated from alternate fiber or plastic sections riveted together to form a scissors-action puller. The center section or sections shall be shaped to form a rib or barrier to provide a firm surface against the fuse body during insertion in fuseholders.

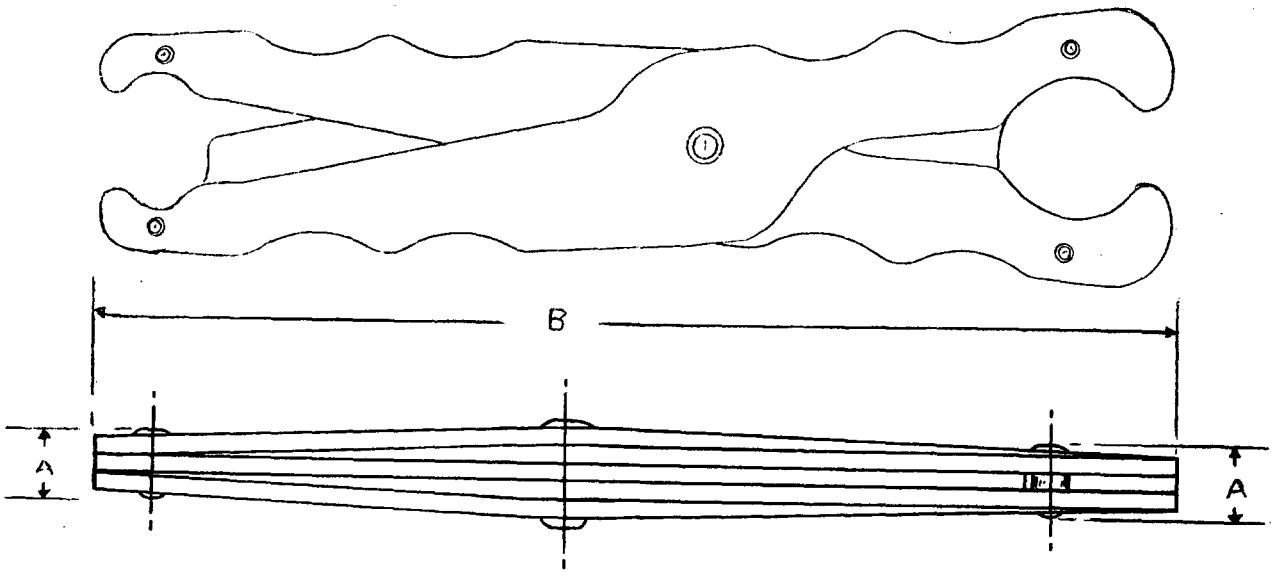


FIGURE 1. Type I, Laminated construction.

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TABLE I. Dimensions

Size no.	1/Jaw thickness inch A	Overall length (inches) ± 1/2 B
1	0.425	4-1/2
2	.500	7-1/2
3	.700	11-1/2

1/Maximum end thickness of layers, including rivets, where applicable.

3.1.2 Type II molded construction type. Type II pullers shall be similar to figure 2, and be of molded plastic construction. The jaw thickness shall not exceed the diameter of the largest fuse to be accommodated. The overall length shall be as specified for type I pullers in table I.

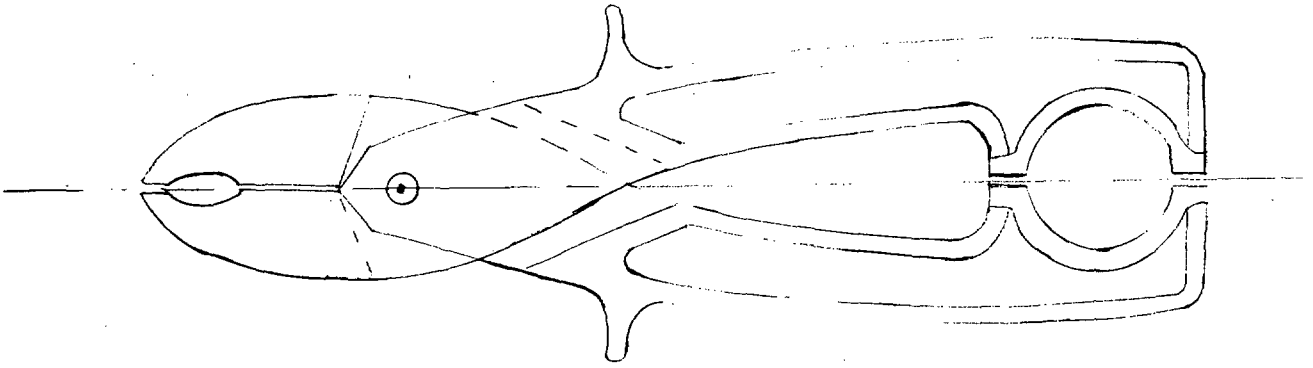


FIGURE 2. Type II, Molded construction.

3.2 Material. Fuse pullers shall be made of laminated sections of sheet vulcanized fiber, or plastic. The material used shall have the necessary characteristics to enable the fuse pullers to pass the tests in 4.3.

3.3 Performance. Fuse pullers shall be required to safely insert and remove fuses of the applicable sizes for the puller from standard fuseholders without becoming disassembled or distorted in use. Operation of the puller shall be smooth and easily accomplished with the use of one hand.

3.3.1 Dielectric strength. The dielectric strength shall be such as to withstand the test specified in 4.3.1 without arcing or visible damage to the fuse puller.

3.3.2 Insulation resistance. The insulation resistance shall be not less than 3 megohms when tested as specified in 4.3.2.

3.3.3 Deflection. Type I pullers shall not deflect more than 0.12 inch when tested as specified in 4.3.3.

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3.4 Identification marking. The manufacturer's name or U. S. registered trademark shall be legibly and permanently marked on the puller in a conspicuous location in addition to any other markings normally included on the equivalent commercial product to indicate intended use or capacity.

3.5 Workmanship. The pullers shall conform to the requirements of this specification and shall be free from any defects which may impair their serviceability or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specifed in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table II.

TABLE II. Sampling data

Category	Sample unit	Inspection level	Acceptable quality level (AQL)	AQL expressed in terms of	Reference
Visual examination	One each	II	2.5	Defects per 100 units	4.2.1
Dimensional examination	One each	S-3	1.5	Defects per 100 units	4.2.2
Testing	One each	S-3	2.5	Defects per 100 units	4.3
Preparation for delivery	One container	S-2	4.0	Defects per 100 units	4.4

4.2 Examination.

4.2.1 Visual examination. Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, and marking.

4.2.2 Dimensional examination. Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.3 Testing.

4.3.1 Dielectric strength. After immersion for 24 hours in a solution containing 20 percent NaCl by weight, the fuse puller shall be washed in fresh water and then allowed to dry for four hours in still air at a room temperature not exceeding 80° F. The puller shall then be subjected to 4,000 volt AC test voltage. This voltage shall be applied for one minute between the points where the fuse is gripped and a point located two inches beyond the pivot of the fuse puller. The requirements of 3.3.1 shall be met during and after this test.

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4.3.2 Insulation resistance. The insulation resistance, specified in 3.3.2, shall be measured between the points where the fuse is gripped and a point located two inches beyond the pivot of the fuse puller, using a megohmmeter (megger) at a direct current potential of 1,000 volts.

4.3.3 Deflection test. The puller shall withstand the deflection test specified on figure 3.

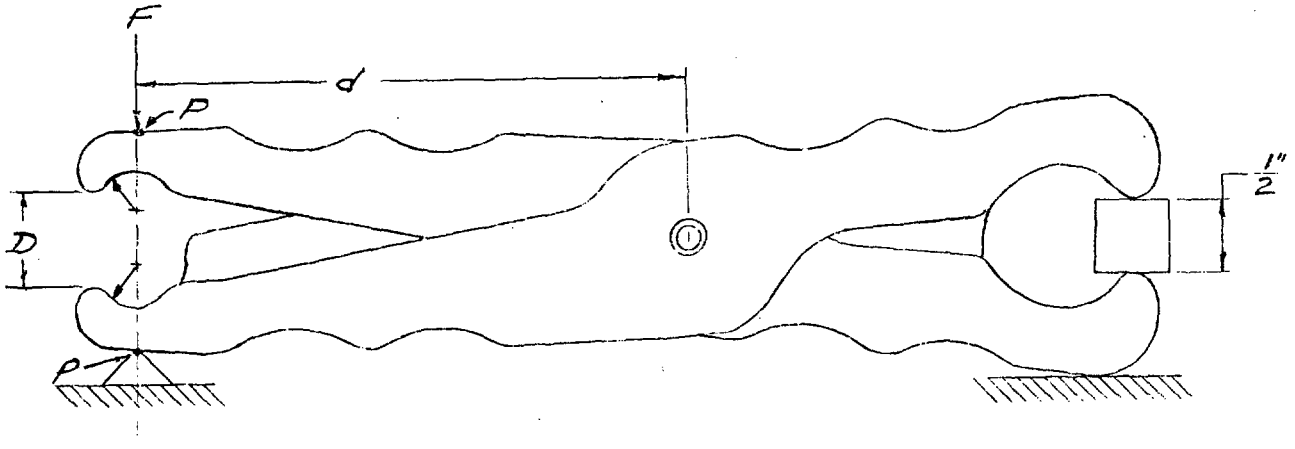


FIGURE 3. Deflection test.

Test to determine deflection at "D" shall be conducted as follows:

(a) Support puller as shown in figure 3 with a 1/2 inch solid bar in location shown. The puller may be notched as required to prevent movement at points "P".

(b) Apply a moment load through points "P" equal to 19.0 inch-pounds. (Example: When d = 3.8 inches, F will equal 19.0/3.8 or 5.0 pounds.)

(c) Measure distance "D" and record as "D_1".

(d) Increase applied moment, as in (b) above, to 57.0 inch-pounds. (Example: When d = 3.8 inches, F will equal 57.0/3.8 or 15 pounds.

(e) Again measure distance "D" and record as "D_2".

(f) The difference between D_1 and D_2 shall not exceed .12 inch.

4.3.4 Mechanical strength and performance. At completion of the tests in 4.3.3, the sample puller shall be used to remove and install a fuse, or dummy fuse, of the proper size for the puller under test. The test shall consist of ten removals and ten insertions. The puller shall meet all the requirements of 3.3.

4.4 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected in accordance with PPP-P-40 to verify compliance with the requirements of section 5.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Fuse pullers shall be preserved, packaged, packed, and marked in accordance with PPP-P-40. Preservation and packaging shall be level A, B, or C, as specified (see 6.1). Packing shall be level A, B, or C, as specified (see 6.1) and marking shall be for civil agencies or military agencies as specified (see 6.2).

6. NOTES

6.1 Intended use. The electrical ratings of the fuses to be pulled by the fuse pullers covered by this specification are as follows:

Type I:
Size 1 - 0-30 ampere, 250 volt.
Size 2 - 0-100 ampere, 250 volt, or
0-100 ampere, 600 volt.
Size 3 - 100-600 ampere, 250 volt, or
60-400 ampere, 600 volt.

Type II:
Size 1 - 0-30 ampere, 250 volt.
Size 2 - 0-100 ampere, 250 volt, or
0-60 ampere, 600 volt.
Size 3 - 110-200 ampere, 250 volt, or
110-200 ampere, 600 volt.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) Selection of applicable level of preservation, packaging, and packing (see 5.1).

6.3 Supersession data. Type I and type II in this revision are replacement designations for class A in W-P-796A. No replacement designation is applicable for class B, but either type I or type II in this specification is an acceptable substitute. The fuse puller sizes in W-P-796A are superseded by the identical sizes for type I fuses in this specification, except that size 4 fuse pullers in W-P-796A are not covered.

MILITARY CUSTODIANS:

DOD has waived coordination on revisions and amendments to this Federal Specification until further notice.

Preparing activity:

GSA-FSS

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein.
Price 10 cents each.

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